VIRASAT ATLAS

DIGITAL HERITAGE ARCHIVE & DISCOVERY PLATFORM

Find the story behind the archive - by subject, place, time and source.

CHALLENGE HC26-CULT01

Digital Revitalisation of Unstructured Cultural & Architectural Heritage Archives

Blithchron '27 | IIT Gandhinagar

Category: Software (Open Innovation) • Domain: Culture, Media & Entertainment

PROPOSED TECHNOLOGY

React.js • PHP REST API • MySQL 8+ • Leaflet / OpenStreetMap

FULL PROJECT REPORT | BUILD SPECIFICATION

Submission deadline: 4 November 2026 • 6:30 PM IST

Report Contents

Executive Summary

1. Challenge Context and Problem Definition
2. Proposed Solution
3. Product Objectives and Beneficiaries
4. Functional Scope
5. User Experience and Key Workflows
6. Technical Architecture
7. Database Design
8. API Design
9. Security, Privacy and Responsible Heritage Practice
10. Scalability and Maintainability
11. Limitations and Mitigations
12. Implementation Roadmap
13. Evaluation Alignment
14. Demonstration Plan
15. Submission Readiness Checklist

Conclusion

Document note: This report is a product and engineering specification. It describes the proposed MVP and roadmap; it does not claim that the application has already been implemented or deployed.

Executive Summary

Virasat Atlas is a map-first digital heritage archive designed to turn scattered photographs, scanned documents, field notes, regional-language material and architectural observations into a structured and discoverable cultural knowledge resource. It responds directly to the HC26-CULT01 challenge: cultural documentation is frequently collected with care but becomes difficult to reuse when it remains distributed across presentation decks, cloud folders and disconnected reports.

The platform provides a governed workflow from contribution to public discovery. Researchers upload source material and describe it with metadata such as title, item type, place, historical period, language, creator, source and rights. An archivist reviews the record before publication. Approved public records become accessible through keyword search, filters, detail pages, a geographic map and curated collections. OCR may assist transcription, but extracted text is stored separately from reviewed text so that machine output is not mistaken for verified research.

The proposed core stack is React.js for the interface, PHP for a versioned REST API and MySQL 8+ for structured records, relationships and full-text search. Leaflet and OpenStreetMap provide geographic exploration. This architecture is intentionally a modular monolith: it is easier to build, test and deploy within a short hackathon window than a collection of microservices, while preserving clear boundaries for later scale.

The MVP is defined around one complete vertical slice: **researcher contribution -> media upload -> metadata and provenance -> review -> publication -> search -> map -> collection**. Automatic 3D reconstruction, vector-based semantic search and broad multilingual translation are future extensions rather than promises of the first prototype. This prioritisation makes the proposal technically credible and gives the judging panel a demonstrable product workflow.

1. Challenge Context and Problem Definition

The challenge brief describes a flow of cultural material produced through grassroots exploration and field documentation, including photographs of heritage structures, regional documents, architectural observations and travel journals. The problem is not necessarily a lack of content; it is the lack of a coherent system to preserve context and make the content usable.

When material is stored in separate formats and folders, a photograph may lose its location, contributor, date, source and interpretation. Text may remain trapped inside scanned documents. Researchers must know which file or folder to open before they can discover useful information. Public visitors cannot easily explore a region's architectural or cultural history as a connected experience. Institutions also lack a consistent workflow to verify content, manage permission and preserve provenance.

1.1 Core problem areas

- **Fragmented storage:** material is distributed across static decks, documents and cloud folders.
- **Poor discoverability:** there is no unified search across descriptions, transcripts, tags and places.
- **Context loss:** files can become detached from location, period, creator and source.
- **Limited public experience:** the archive is not easily navigable as a map, collection or cultural story.
- **Weak governance:** rights, attribution, transcription corrections and publication decisions are difficult to track consistently.

1.2 Opportunity

Treat every asset as a cultural record with relationships, not simply as a file. A single record can be discovered by its subject, location, period, language, tag, collection or source. That creates an archive that can support research as well as public exploration.

2. Proposed Solution

Virasat Atlas is a unified archive and discovery platform with four connected capabilities:

1. **Structured ingestion:** upload a photograph, document or field note and attach meaningful metadata.
2. **Responsible enrichment:** add tags, location, source, creator, dates, rights and, where suitable, an OCR-assisted transcript.
3. **Editorial review:** verify essential metadata and rights before a record becomes public.
4. **Multi-dimensional discovery:** search, filter, browse a map, inspect detail pages and follow curated collections.

2.1 Product positioning

Virasat Atlas is not only a gallery, document repository or tourism map. It is a cultural archive that reconnects an asset to the people, place, time, source and interpretation that make it meaningful.

2.2 Primary product promise

Find the story behind the archive - by subject, place, time and source.

3. Product Objectives and Beneficiaries

3.1 Objectives

- Consolidate heterogeneous heritage material into a searchable catalogue.
- Preserve provenance, rights and contextual metadata with each item.
- Make geography a first-class discovery dimension.
- Provide a review workflow that separates contribution from publication.
- Support transcription assistance without treating OCR output as fact.
- Keep the implementation achievable with React, PHP and MySQL.

3.2 Beneficiaries

Beneficiary	Value delivered
Cultural researchers	Searchable records with source, language, place and context
Academic institutions	A repeatable archive workflow and structured metadata
Digital archivists	Review controls, provenance, rights and correction history
Heritage tourism sector	Place-based discovery and curated heritage collections
Students and public visitors	Accessible exploration through search, maps and stories
Field contributors	A consistent place to submit photographs and observations

4. Functional Scope

4.1 Researcher workspace

Researchers authenticate, create an archive record, upload supporting files, add metadata and save a draft or submit it for review. The contribution form is organised into record, place/time, source, media, rights and transcript sections. Upload limits and accepted file types are visible before submission.

4.2 Archive record model

A record includes a title, item type, summary, description, language, place, coordinates when known, historical period, capture date, creator, source, tags, rights status and attribution. Media assets are linked to records, but stored outside the database as files. This keeps MySQL focused on metadata and relationships.

4.3 Review and publication

Records move through draft, pending_review, published, rejected and archived states. Visibility is independently controlled as public, unlisted or private. Anonymous discovery endpoints return only records that are both published and public. Reviewers can approve, reject or return a submission with a reason. Rights and attribution fields are checked before approval.

4.4 Search and filters

The MVP uses MySQL FULLTEXT search across title, summary, description and reviewed transcript text. Filters include item type, region, language, period and tags. Results are paginated by the backend; the browser does not download the entire archive. Search results link to the same canonical item record used by the map and collections.

4.5 Map exploration

A Leaflet map displays published, public records with valid coordinates. Selecting a marker opens an item preview and then the canonical detail page. A list view remains available as an accessible alternative and as a fallback when map tiles are unavailable. Approximate coordinates are labelled rather than presented as exact.

4.6 Collections and heritage trails

Curators group items into a collection based on theme, location, craft, architecture or a field-study route. Each collection has a title, summary, context note, curator and an ordered list of archive records. Route lines are only displayed when the sequence and coordinates are meaningful; a curated order must not imply safe public access where that has not been verified.

4.7 OCR-assisted transcription

OCR can assist with digitising supported scans. The workflow records the engine, language and raw extracted text. A researcher or reviewer checks the output against the source and saves a reviewed transcript. Only reviewed text is indexed for normal discovery by default. Language support must be tested against representative documents; historical handwriting and degraded scans may require manual transcription.

5. User Experience and Key Workflows

5.1 Researcher contribution

1. Sign in.
2. Start a new archive record.
3. Enter title, type and summary.
4. Add place, date/period, language and contributor/source.
5. Upload a file and add caption/alt text.
6. Set rights status and attribution.
7. Add tags and optional transcript.
8. Save draft or submit for review.

5.2 Reviewer moderation

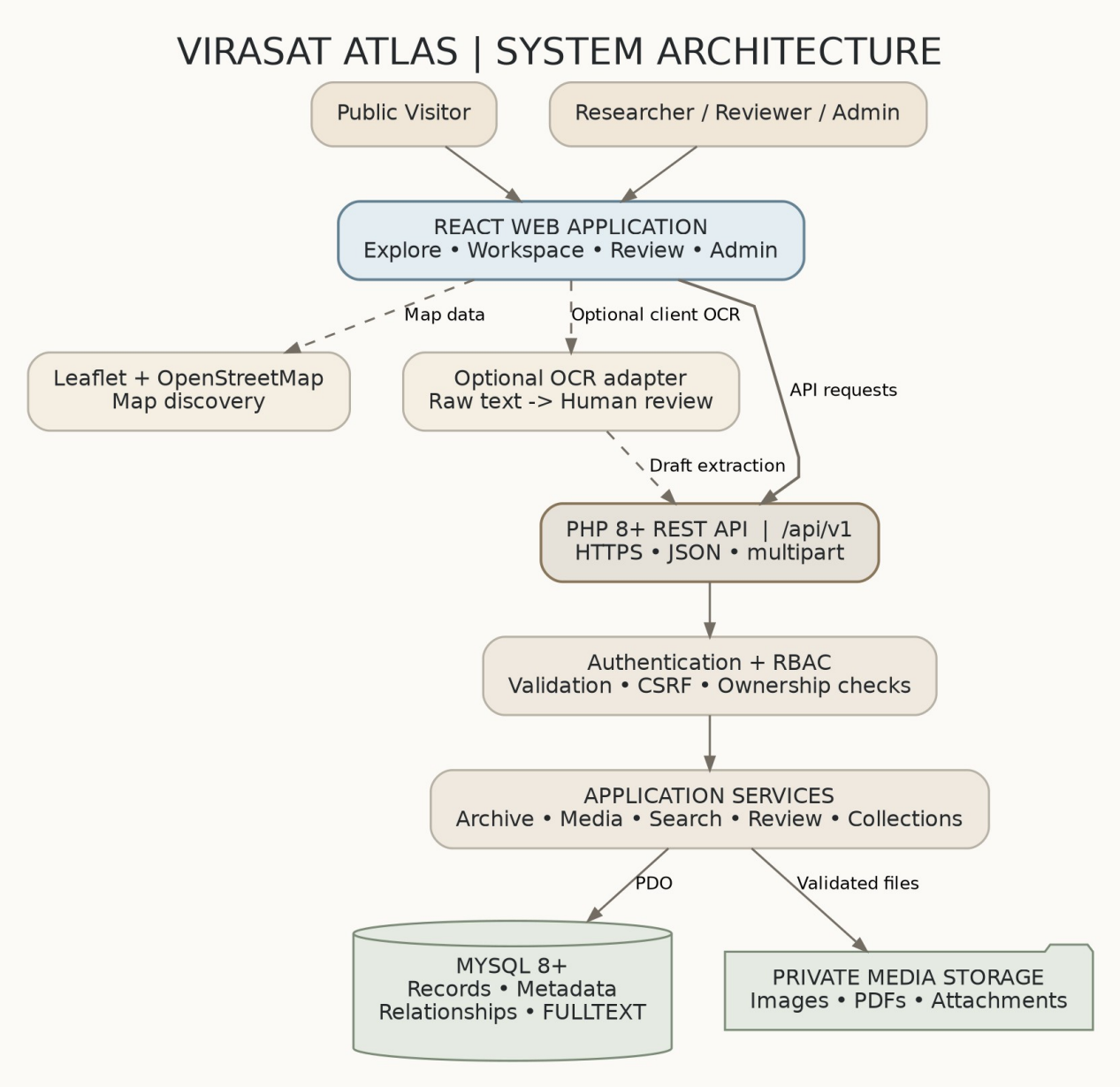
1. Open the pending review queue.
2. Inspect the source file and metadata.
3. Verify source and rights information.
4. Correct metadata and reviewed transcript if required.
5. Approve, reject or return for correction with a reason.
6. Publish the record subject to its visibility setting.

5.3 Public discovery

1. Search by a subject or place.
2. Refine by region, type, language or period.
3. Select a result or explore the map.
4. Inspect images, narrative, source, rights and location.
5. Open a related collection and browse its ordered records.

6. Technical Architecture

6.1 Architecture pattern



System architecture

A modular monolith is proposed for the MVP. React handles presentation and interactions. PHP owns application rules and exposes a versioned REST API. MySQL stores the structured archive and indexed search fields. Media files are stored separately from MySQL in a private storage directory.

6.2 Component responsibilities

Component	Responsibility
React frontend	Public discovery, contributor forms, review UI and map
PHP API	Authentication, validation, authorisation, business rules and file handling

Component	Responsibility
MySQL	Users, archive records, locations, tags, media metadata, collections, OCR jobs and audit events
Private media store	Original images and documents; access controlled by API
Leaflet	Map rendering and marker interaction
OCR adapter	Optional extraction; returns draft text for review

6.3 Why this stack

- React supports reusable page and component patterns for the public catalogue and staff workspaces.
- PHP is a practical backend for the required stack and can expose REST endpoints without another server runtime.
- MySQL supports relational integrity, joins, indexes and FULLTEXT search for the initial catalogue.
- Leaflet provides map interaction without requiring a paid map SDK for the prototype.
- Keeping OCR behind an adapter prevents an experimental engine from becoming a core dependency.

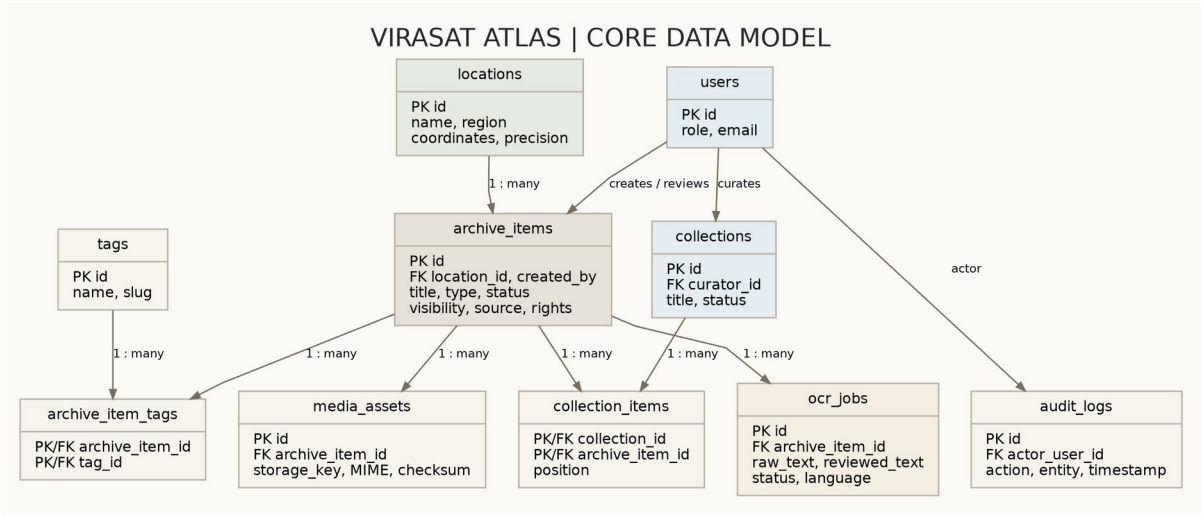
6.4 Suggested repository structure

```
client/                  # React + Vite
server/                  # PHP REST API
  public/index.php           # API front controller
  src/Controllers/
  src/Services/
  src/Repositories/
  src/Middleware/
  src/Validators/
  storage/                 # private files
database/schema.sql        # MySQL schema
docs/                    # diagrams and implementation documents
```

7. Database Design

The schema is relational and keeps a clear separation between a cultural record and its binary attachments.

7.1 Core entities



Core entity relationship diagram

Entity	Key attributes	Purpose
users	id, name, email, password_hash, role	Authentication and permissions
locations	id, name, region, latitude, longitude, precision	Geographic context
archive_items	id, slug, title, type, summary, description, status, visibility, source, rights, location_id	Canonical cultural record
media_assets	id, archive_item_id, storage_key, MIME, size, checksum, alt_text	File references and metadata
tags	id, name, slug	Controlled vocabulary
archive_item_tags	archive_item_id, tag_id	Many-to-many tagging
collections	id, title, curator, status, visibility	Curated themes or trails
collection_items	collection_id, archive_item_id, position	Ordered collection membership
ocr_jobs	id, item_id, raw_text, reviewed_text, language, status	Transcription workflow
audit_logs	actor, action, entity, timestamp	Accountability for key actions

7.2 Relationship summary

- One user can create many archive items.
- One location can be associated with many archive items.
- One archive item can have many media assets and many tags.
- Tags can be shared by many archive items.
- Collections contain many archive items, and an item can appear in multiple collections.
- An archive item can have multiple OCR attempts, each preserving raw and reviewed text separately.

7.3 Data integrity

Foreign keys preserve relationships. utf8mb4 supports multilingual text storage. Full-text indexes support catalogue search. Visibility and review state are separate to support public, unlisted and private records. Coordinates are optional and include a precision label so approximate information is not presented as exact.

The implementation-ready SQL is supplied in database/schema.sql.

8. API Design

All API endpoints are versioned under /api/v1. The API returns a consistent JSON envelope with data, optional pagination meta and error.

Method	Endpoint	Purpose
POST	/auth/login	Authenticate user
POST	/auth/logout	End session
GET	/auth/me	Current user and role
GET	/archive-items	Search and paginate public records
POST	/archive-items	Create draft
GET	/archive-items/{id}	Read accessible record
PATCH	/archive-items/{id}	Update permitted fields
POST	/archive-items/{id}/media	Attach media
GET	/locations	Return map data
GET	/collections	List public collections
POST	/collections	Create collection
GET	/review-queue	Retrieve submissions awaiting review
POST	/archive-items/{id}/review	Approve/reject/return record
POST	/archive-items/{id}/ocr-jobs	Register OCR result
PATCH	/ocr-jobs/{id}	Save reviewed transcript

9. Security, Privacy and Responsible Heritage Practice

9.1 Application security

- Passwords are hashed with PHP password_hash() and verified with password_verify().
- PDO prepared statements are used for database queries.
- Role and ownership checks are enforced in PHP for every protected operation.
- Session cookies use HttpOnly, Secure in HTTPS and appropriate SameSite settings; CSRF protection is applied to write requests.
- CORS is restricted to exact allowed origins when cross-origin deployment is required.
- Uploads are size/type checked and assigned random server-side filenames.
- Original media is stored outside the public web root.

- Public endpoints exclude draft, private and non-public records.

9.2 Heritage data responsibility

Cultural archives can contain sensitive community knowledge, restricted sites or material with unclear permissions. The platform therefore records provenance and rights status and supports restricted visibility. Where rights are unknown, the item should not be made publicly downloadable. Contributors' attribution preferences must be respected. Site coordinates may need to be generalised or withheld when precise disclosure could create a risk of damage, theft or unauthorised access.

9.3 Research integrity

The system distinguishes capture date from historical period, source fact from interpretation, and machine transcription from reviewed text. Uncertain details should be labelled as uncertain. Publication status is a moderation decision, not a claim that every historical interpretation has been independently verified.

10. Scalability and Maintainability

The initial platform is sized for a prototype and early institutional archive. It avoids premature distributed infrastructure but preserves clear service boundaries.

Near-term growth

- Generate thumbnails and previews asynchronously.
- Move files from local storage to S3-compatible object storage.
- Add a queue for OCR, document conversion, image processing and virus scanning.
- Cache public catalogue responses after observing real traffic.
- Add monitoring for API latency, upload failures, search performance and missing media.

Longer-term growth

- Add a vector index for semantic search after a multilingual evaluation set is created.
- Add dedicated geospatial query infrastructure if MySQL query plans require it.
- Add translation review and language-specific OCR pipelines.
- Add IIIF-compatible image delivery or institutional metadata export if partner archives need it.
- Add a 3D model viewer for assets that have been reconstructed and validated using a separate photogrammetry workflow.

11. Limitations and Mitigations

Limitation	Why it matters	Mitigation / next step
OCR quality varies	Historical scans and regional scripts may be difficult to recognise	Language-specific testing and human review
MySQL FULLTEXT is not full semantic search	Synonyms and multilingual intent may be missed	Improve tags/synonyms first; evaluate vector search later
2D records do not provide a 3D digital twin	The 3D pathway is technically demanding	Link validated models in a future WebGL/Three.js module
Local file storage is not ideal at scale	Backups and distribution become harder	Move to object storage and CDN
Map accuracy depends on source data	Incorrect coordinates mislead users	Label approximate coordinates and allow reviewer correction

Limitation	Why it matters	Mitigation / next step
Rights data may be incomplete	Public reuse could be inappropriate	Restrict visibility until permission is clear
Third-party map tiles require connectivity	Map may not render offline	Keep a list-based discovery fallback

12. Implementation Roadmap

Hackathon MVP

- Build the React application, PHP API and MySQL schema.
- Implement authentication and role-based access.
- Create contribution, upload and review workflow.
- Implement search, filters, detail pages, map and collections.
- Seed a small set of rights-cleared example records.
- Deploy, document and demonstrate the end-to-end flow.

Phase 2 - Research enrichment

- Add OCR jobs with language selection and review UI.
- Add richer PDF/image preview and thumbnail generation.
- Add improved taxonomy and related-record recommendations.
- Add audit-log interface and archive export.

Phase 3 - Immersive and semantic capabilities

- Add semantic search after creating a relevance benchmark.
- Add reviewed multilingual translations.
- Integrate validated 3D assets with Three.js.
- Add map-based story routes and institution-level collections.

13. Evaluation Alignment

The supplied evaluation weights are Technical Execution & Code Quality (40%), System Architecture & Scalability (30%), UI/UX (20%) and Innovation & Problem Fit (10%).

Evaluation area	Virasat Atlas response	Evidence to show
Technical execution & code quality (40%)	Real upload, validation, DB persistence, API and review workflow	Live contribution-to-publication demo; clean repository and schema
Architecture & scalability (30%)	Modular PHP API, relational schema, private media storage, versioned endpoints	Architecture diagram, ER model, security boundaries and growth path
UI/UX (20%)	Editorial archive experience, search filters, detail pages, map/list and accessible forms	Live desktop/mobile walkthrough and map/detail interaction
Innovation & problem fit (10%)	Connects archive records to place, provenance and curated cultural journeys	Demonstrate same item through search, map and collection; OCR review path

14. Demonstration Plan (2-3 minutes)

0:00-0:20 - Problem and concept: introduce Virasat Atlas as a structured archive that connects media to source, location and context. **0:20-0:55 - Contribution:** sign in as researcher, create a record and upload a real sample file. **0:55-1:20 - Review:** open the reviewer queue and approve the record after checking metadata and rights. **1:20-2:05 - Discovery:** search for the item, open its detail page and show provenance/attribution. **2:05-2:35 - Map and collections:** show the record on the map and open its curated collection. **2:35-2:50 - Roadmap:** explain OCR review, semantic search and validated 3D models as future extensions.

The recording must show the working application, not a slide-only simulation. Any seeded or pre-populated data should be identified honestly.

15. Submission Readiness Checklist

- Public GitHub/GitLab repository with relevant hackathon-window commits.
- Comprehensive README with architecture diagrams and clean setup steps.
- Reproducible MySQL schema and sample data that is cleared for demonstration.
- Working prototype deployed and smoke-tested.
- Unlisted YouTube/Loom video of 2-3 minutes showing real features.
- Accessible pitch deck PDF with no more than 10 slides.
- Pitch deck includes problem, architecture, database schema, tech stack, scalability, limitations and roadmap.
- Submission completed before 4 November 2026, 6:30 PM IST.

Conclusion

Virasat Atlas addresses the challenge at the point where cultural value is most often lost: between collecting a record and being able to find, interpret and responsibly reuse it. By connecting media with metadata, provenance, review, geography and curated collections, it turns disconnected files into a navigable archive. Its React + PHP + MySQL architecture is achievable for a hackathon MVP and leaves clear extension points for OCR, semantic retrieval and immersive 3D experiences. The strongest proof of the idea is a complete record journey - from researcher upload to public discovery across search, map and collections.